Lecture 6: Linear Maps

MATH 110-3

Franny Dean

June 28, 2023

Context

We started with vector spaces.

Context

We started with vector spaces.

> "No one gets excited about vector spaces. The interesting part of linear algebra is the subject to which we now turn – linear maps."
> - *S. Axler*

Definition

Let V, W be $\mathbb{F}$ vector spaces. ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$).

Def'n:

A **linear map** from V to W is a function $T : V \to W$ satisfying the following properties:

Definition

Let V, W be $\mathbb{F}$ vector spaces. ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$).

Def'n:

A **linear map** from V to W is a function $T : V \to W$ satisfying the following properties:

Additivity:

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in V$$

Definition

Let V, W be $\mathbb{F}$ vector spaces. ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$).

Def'n:

A **linear map** from V to W is a function $T : V \to W$ satisfying the following properties:

Additivity:

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in V$$

Homogeneity:

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in V, \lambda \in \mathbb{F}$$

Definition

Let V, W be $\mathbb{F}$ vector spaces. ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$).

Def'n:

A **linear map** from V to W is a function $T : V \to W$ satisfying the following properties:

Additivity:

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in V$$

Homogeneity:

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in V, \lambda \in \mathbb{F}$$

Def'n:

The set of all linear maps V to W is denoted $\mathcal{L}(V, W)$.

Examples!

- The zero map. $T(v) = 0$.

Examples!

- The zero map. $T(v) = 0$.
- The identity map. $I(v) = v$.

Examples!

- The zero map. $T(v) = 0$.
- The identity map. $I(v) = v$.
- Differentiation. $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ defined $Dp = p'$.

Examples!

- The zero map. $T(v) = 0$.
- The identity map. $I(v) = v$.
- Differentiation. $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ defined $Dp = p'$.
- Integration. $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined $Tp = \int_0^1 p dx$.

Examples!

- The zero map. $T(v) = 0$.
- The identity map. $I(v) = v$.
- Differentiation. $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ defined $Dp = p'$.
- Integration. $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined $Tp = \int_0^1 p dx$.
- From $\mathbb{R}^3 \to \mathbb{R}^2$: $T(x, y, z) = (2x - y, 5x - 7z, 2z - y)$.

Examples!

- The zero map. $T(v) = 0$.
- The identity map. $I(v) = v$.
- Differentiation. $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ defined $Dp = p'$.
- Integration. $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined $Tp = \int_0^1 p dx$.
- From $\mathbb{R}^3 \to \mathbb{R}^2$: $T(x, y, z) = (2x - y, 5x - 7z, 2z - y)$.
- Generalizing from $\mathbb{F}^n \to \mathbb{F}^m$:
  Let $A_{j,k} \in \mathbb{F}$ for $j \in [m]$, $k \in [n]$, define $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ as

$$T(x_1, \ldots, x_n) = (A_{1,1}x_1 + \ldots + A_{1,n}x_n, \ldots, A_{m,1}x_1 + \ldots + A_{m,n}x_n).$$

Basis of Domain

Prop'n 3.5:

Suppose $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_j = w_j$$

for each $j = 1, \ldots, n$.

Basis of Domain

Prop'n 3.5:

Suppose $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_j = w_j$$

for each $j = 1, \ldots, n$.

What does this mean?

Basis of Domain

Prop'n 3.5:

Suppose $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_j = w_j$$

for each $j = 1, \ldots, n$.

What does this mean?

1. I can give you any basis of V and any vectors of W and create a map sending basis vectors to these vectors of W.

Basis of Domain

Prop'n 3.5:

Suppose $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_j = w_j$$

for each $j = 1, \ldots, n$.

What does this mean?

1. I can give you any basis of V and any vectors of W and create a map sending basis vectors to these vectors of W.
2. A linear map is uniquely determined by where it sends basis vectors.

Proof of Prop'n 3.5

WTS:

1. Existence
2. Uniqueness

Proof of Prop'n 3.5

WTS:

1. Existence
2. Uniqueness

Existence:
Define $T : V \to W$ by

$$T(c_1 v_1 + \ldots + c_n v_n) = c_1 w_1 + \ldots + c_n w_n$$

for arbitrary $c_1, \ldots, c_n \in \mathbb{F}$.

Proof of Prop'n 3.5

WTS:

1. Existence
2. Uniqueness

Existence:
Define $T : V \to W$ by

$$T(c_1 v_1 + \ldots + c_n v_n) = c_1 w_1 + \ldots + c_n w_n$$

for arbitrary $c_1, \ldots, c_n \in \mathbb{F}$.

For each j, if $c_j = 1$ and all other $c_i = 0$, then $Tv_j = w_j$.

Proof of Prop'n 3.5

WTS:

1. Existence
2. Uniqueness

Existence:
Define $T : V \to W$ by

$$T(c_1 v_1 + \ldots + c_n v_n) = c_1 w_1 + \ldots + c_n w_n$$

for arbitrary $c_1, \ldots, c_n \in \mathbb{F}$.

For each j, if $c_j = 1$ and all other $c_i = 0$, then $Tv_j = w_j$.

Proof of Prop'n 3.5 (Cont'd)

To show T is linear:

Proof of Prop'n 3.5 (Cont'd)

To show T is linear:
Let $x = a_1v_1 + \ldots + a_nv_n, y = b_1v_1 + \ldots + b_nv_n$.

$$\begin{aligned} T(x+y) &= T((a_1+b_1)v_1 + \ldots (a_n+b_n)v_n) \\ &= (a_1+b_1)w_1 + \ldots (a_n+b_n)w_n \\ &= (a_1w_1 + \ldots a_nw_n) + (b_1w_1 + \ldots b_nw_n) \\ &= Tx + Ty \end{aligned}$$

Proof of Prop'n 3.5 (Cont'd)

To show T is linear:
Let $x = a_1v_1 + \ldots + a_nv_n, y = b_1v_1 + \ldots + b_nv_n$.

$$\begin{aligned} T(x+y) &= T((a_1+b_1)v_1 + \ldots (a_n+b_n)v_n) \\ &= (a_1+b_1)w_1 + \ldots (a_n+b_n)w_n \\ &= (a_1w_1 + \ldots a_nw_n) + (b_1w_1 + \ldots b_nw_n) \\ &= Tx + Ty \end{aligned}$$

Scaling left to you!

Proof of Prop'n 3.5 (Cont'd)

To show T is linear:
Let $x = a_1v_1 + \ldots + a_nv_n, y = b_1v_1 + \ldots + b_nv_n$.

$$\begin{aligned} T(x+y) &= T((a_1+b_1)v_1 + \ldots (a_n+b_n)v_n) \\ &= (a_1+b_1)w_1 + \ldots (a_n+b_n)w_n \\ &= (a_1w_1 + \ldots a_nw_n) + (b_1w_1 + \ldots b_nw_n) \\ &= Tx + Ty \end{aligned}$$

Scaling left to you!

Done with **existence!**

Proof of Prop'n 3.5 (Cont'd)

Uniqueness:

Proof of Prop'n 3.5 (Cont'd)

Uniqueness:

S'pose $Tv_j = w_j$ for some linear map T.

Proof of Prop'n 3.5 (Cont'd)

Uniqueness:

S'pose $Tv_j = w_j$ for some linear map T.

Homogeneity implies $T(c_j v_j) = c_j w_j$.

Proof of Prop'n 3.5 (Cont'd)

Uniqueness:

S'pose $Tv_j = w_j$ for some linear map T.

Homogeneity implies $T(c_j v_j) = c_j w_j$.

Then additivity implies $T(c_1 v_1 + \ldots + c_n v_n) = c_1 w_1 + \ldots + c_n w_n$.

Proof of Prop'n 3.5 (Cont'd)

Uniqueness:

S'pose $Tv_j = w_j$ for some linear map T.

Homogeneity implies $T(c_j v_j) = c_j w_j$.

Then additivity implies $T(c_1 v_1 + \ldots + c_n v_n) = c_1 w_1 + \ldots + c_n w_n$.

Thus, T must be the map we just defined.

Algebra of $\mathcal{L}(V, W)$

Def'n:

Let $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. We define for all $v \in V$

$$(S + T)(v) = Sv + Tv$$

and

$$(\lambda T)(v) = \lambda T(v).$$

Algebra of $\mathcal{L}(V, W)$

Def'n:

Let $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. We define for all $v \in V$

$$(S + T)(v) = Sv + Tv$$

and

$$(\lambda T)(v) = \lambda T(v).$$

Prop'n:

With the addition and scalar multiplication defined above, $\mathcal{L}(V, W)$ is a **vector space**.

More algebra of $\mathcal{L}(V, W)$

Def'n:

Let U, V, W be vector spaces. If $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$, then we define $ST \in \mathcal{L}(U, W)$ for $u \in U$ as

$$(ST)(u) = S(Tu).$$

More algebra of $\mathcal{L}(V, W)$

Def'n:

Let U, V, W be vector spaces. If $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$, then we define $ST \in \mathcal{L}(U, W)$ for $u \in U$ as

$$(ST)(u) = S(Tu).$$

Example: Let $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be differentiation.

More algebra of $\mathcal{L}(V, W)$

Def'n:

Let U, V, W be vector spaces. If $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$, then we define $ST \in \mathcal{L}(U, W)$ for $u \in U$ as

$$(ST)(u) = S(Tu).$$

Example: Let $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be differentiation.

Let $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be multiplication by x^2.

$$TD(p(x)) = T(p'(x)) = x^2 p'(x)$$

More algebra of $\mathcal{L}(V, W)$

Def'n:

Let U, V, W be vector spaces. If $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$, then we define $ST \in \mathcal{L}(U, W)$ for $u \in U$ as

$$(ST)(u) = S(Tu).$$

Example: Let $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be differentiation.

Let $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be multiplication by x^2.

$$TD(p(x)) = T(p'(x)) = x^2 p'(x)$$

$$DT(p(x)) = D(x^2 p(x)) = 2xp(x) + x^2 p'(x)$$

More algebra of $\mathcal{L}(V, W)$

Def'n:

Let U, V, W be vector spaces. If $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$, then we define $ST \in \mathcal{L}(U, W)$ for $u \in U$ as

$$(ST)(u) = S(Tu).$$

Example: Let $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be differentiation.

Let $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathcal{P}(\mathbb{R}))$ be multiplication by x^2.

$$TD(p(x)) = T(p'(x)) = x^2 p'(x)$$

$$DT(p(x)) = D(x^2 p(x)) = 2xp(x) + x^2 p'(x)$$

⚠ Notice that $TD \neq DT$.

Properties of map composition

Prop'n 3.9:

Properties of map composition

Prop'n 3.9:

- **associativity**

$$(T_1 T_2) T_3 = T_1 (T_2 T_3)$$

*when domains are appropriate

Properties of map composition

Prop'n 3.9:

- **associativity**

$$(T_1T_2)T_3 = T_1(T_2T_3)$$

*when domains are appropriate

- **identity:**

$$TI_V = I_WT = T$$

for $T \in \mathcal{L}(V, W)$

Properties of map composition

Prop'n 3.9:

- **associativity**

$$(T_1T_2)T_3 = T_1(T_2T_3)$$

*when domains are appropriate

- **identity:**

$$TI_V = I_WT = T$$

for $T \in \mathcal{L}(V, W)$

- **distributive properties:**

$$(S_1 + S_2)T = S_1T + S_2T$$

$$S(T_1 + T_2) = ST_1 + ST_2$$

where $T's \in \mathcal{L}(U, V)$ and $S's \in \mathcal{L}(V, W)$

Break

Discussion Questions

1. Prove that all linear maps send 0 to 0.
2. Describe the subspaces of $\mathbb{R}^3$ and their dimensions.
3. Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p''(6) = 0\}$.
   (a) Find a basis of U.
   (b) Extend your basis to one of $\mathcal{P}_4(\mathbb{R})$.
   (c) Find a subspace W such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.
4. Suppose $b, c \in \mathbb{R}$. Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz)$. Show that T is linear if and only if $b = c = 0$.
5. Construct a linear map $T : \mathbb{R}^3 \to \mathbb{R}^2$ such that $T(3, 0, 2) = (3, 4)$ and $T(1, 1, 1) = (1, 1)$. Or show why this is impossible. what about one such that $T(3, 0, 2) = (3, 4)$ and $T(6, 0, 4) = (1, 1)$?

References

[Axl14] Sheldon Axler.
Linear Algebra Done Right.
Undergraduate Texts in Mathematics. Springer Cham, 2014.